

CHAPTER 9

Foreign Exchange Rate Determination

Eiteman, Stonehill & Moffett · Multinational Business Finance, 14e

"The herd instinct among forecasters makes sheep look like independent thinkers." — Edgar R. Fiedler

Learning Objectives

LO1

Explore the three major theoretical approaches to exchange rate determination

LO2

Detail how direct and indirect foreign exchange market intervention is conducted by central banks

LO3

Analyze the primary causes of exchange rate disequilibrium in emerging market currencies

LO4

Observe how forecasters combine technical analysis with the three major theoretical approaches

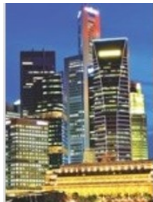
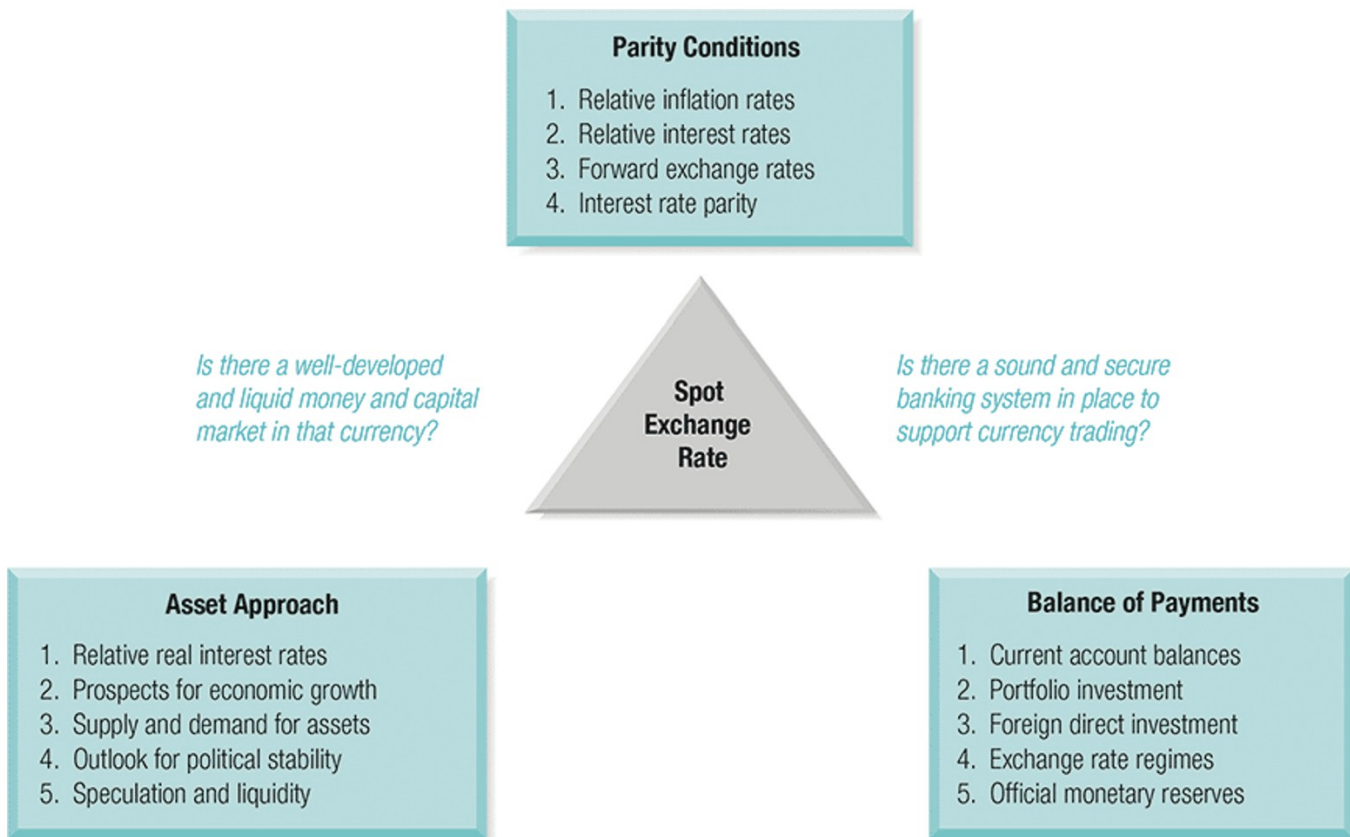


Exhibit 9.1 The Determinants of Foreign Exchange Rates



Purchasing Power Parity (PPP) Approach

What PPP Says

- ▶ The oldest and most widely followed of all exchange rate determination theories.
- ▶ States that the long-run equilibrium exchange rate is determined by the ratio of domestic prices relative to foreign prices.
- ▶ Most theories of exchange rate determination have PPP elements embedded within their frameworks.
- ▶ Three variants: Law of One Price → Absolute PPP → Relative PPP (most practically relevant).

Relative PPP Formula

$$S_1 = S_0 \times (1 + \pi_d) / (1 + \pi_f)$$

S = spot rate π = inflation rate d = domestic f = foreign

⚠ Why PPP Falls Short

Poor short-to-medium term forecasting. Only relative prices matter here — but investment incentives, economic growth, and political change also drive currencies.

Worked Example

Current spot rate: ¥90.00/\$ | Japan inflation: 2% | US inflation: 1%

→ $S_1 = ¥90.00 \times (1.02 / 1.01) = ¥90.89/\$$ (yen depreciates slightly — higher inflation country's currency weakens)

Three Complementary Schools of Thought (Exhibit 9.1)

These are not competing theories — they are complementary. Together they capture the full complexity of exchange rate determination. Two additional institutional dimensions matter: the development of capital markets and banking systems.

Parity Conditions

(PPP-Based)

Oldest & most widely followed

- ▶ Relative inflation rates
- ▶ Relative interest rates
- ▶ Forward exchange rates
- ▶ Interest rate parity

Balance of Payments

(Flow-Based)

Supply & demand for currencies

- ▶ Current account balances
- ▶ Portfolio investment
- ▶ Foreign direct investment
- ▶ Exchange rate regimes
- ▶ Official monetary reserves

Asset Market

(Stock-Based)

Supply & demand for financial assets

- ▶ Relative real interest rates
- ▶ Prospects for economic growth
- ▶ Supply & demand for assets
- ▶ Outlook for political stability
- ▶ Speculation & liquidity

★ *Most determinants are mutually determined — the spot rate both affects and is affected by each of them.*

The Monetary Approach to Exchange Rates

Core idea: Exchange rate is determined by the supply and demand for national **monetary stocks**. Both domestic and foreign bonds are viewed as perfect substitutes — only **money matters**.

How It Works

1. Exchange rate = supply & demand for national monetary stocks.
2. Also depends on expected future levels and growth rates of monetary stocks.
3. Focuses on changes in money supply as the primary driver of inflation.
4. Changes in inflation rates then alter exchange rates via a PPP effect.
5. Assumes prices are flexible in both short and long run → immediate transmission of inflation.

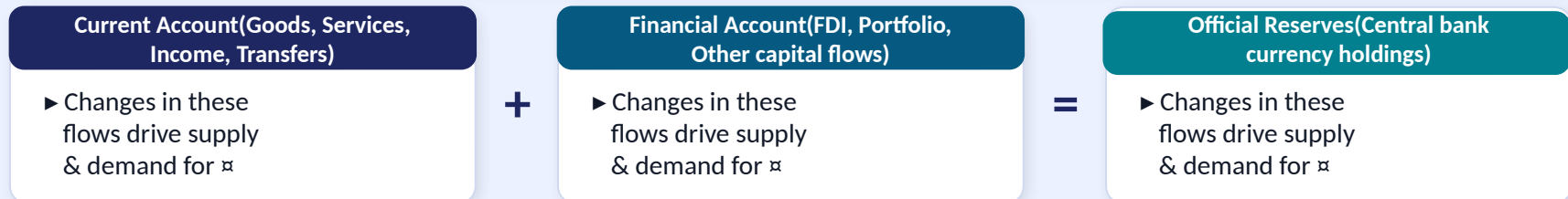
Key Weaknesses

1. Real economic activity only influences exchange rates through changes in money demand — too narrow.
2. Failure of PPP to hold in the short-to-medium term undermines the monetary approach.
3. Money demand appears relatively unstable over time.
4. Level of economic activity and money supply appear interdependent, not independent.
5. Non-monetary assets (bonds, equities) play no role — a significant omission.

Bottom line: The monetary approach is theoretically important but empirically limited. Its legacy lives on in the Asset Market Approach.

Balance of Payments (BOP) Flows Approach

Core idea: Equilibrium exchange rate is found when the net inflow of foreign exchange from **current account activities** matches the net outflow from **financial account activities**.



→ **Equilibrium: inflows = outflows** ⇒ **Exchange rate is at BOP equilibrium**

✓ Why It's Widely Used

- Data widely reported by statistical agencies
- Most frequently captured economic transactions
- Popular with currency traders and business press
- Intuitive: trade flows → currency demand

⚠ Academic Criticism

- Focuses on flows, not stocks of money/financial assets
- Stocks of money and bonds play no role (a key weakness)
- Balance of payments approach largely dismissed by academia
- Practitioners still rely on BOP variations heavily

Asset Market Approach (Portfolio Balance)

Core idea: Exchange rates are determined by supply and demand for a wide variety of **financial assets**. Shifts in supply/demand for assets alter exchange rates. Changes in monetary and fiscal policy alter expected returns and perceived relative risks.

Key Insight from Bruno Solnik: Contemporaneous international flows should have little effect on exchange rates to the extent they have already been expected. Only news about future economic prospects will affect exchange rates.



Relative Real Interest Rates

Major factor for developed markets



Economic Growth & Profitability

Cross-border equity & FDI driver



Supply & Demand for Assets

Stocks, bonds, money market



Political Stability

Risk premium on securities



Capital Market Liquidity

Ease of buying AND selling



Corporate Governance

Shareholder wealth focus



Contagion Effects

Spread of crisis across borders



Speculation

Can trigger or worsen crises

Technical Analysis & the JPY/USD Rate (GFP 9.1)

Definition: Technical analysis is the belief that studying past price behavior provides insights into future price movements.

Key assumption: exchange rates, like all market-driven prices, follow trends that can be analyzed and projected.

Day-to-Day Movement

Seemingly random. Provides little forecasting signal. Best modeled as a random walk (<30 days).

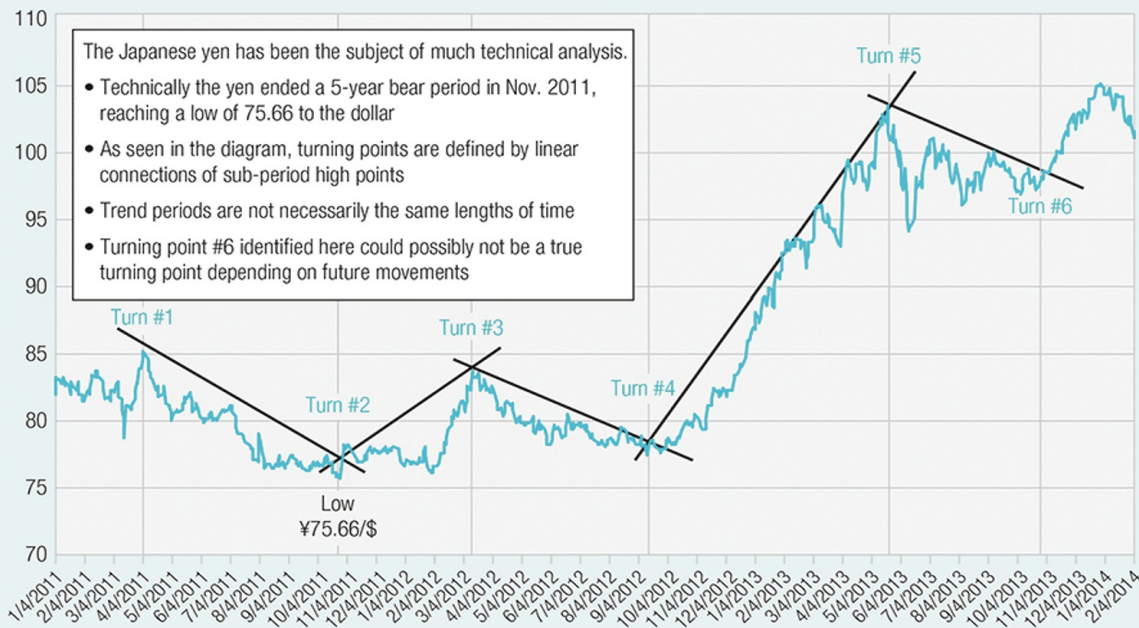
Short-Term Movement

Days to several months. Trends emerge that can be detected and projected using technical indicators.

Long-Term Movement

Characterized by up and down long-term trends. New research suggests 'technical waves' exist under floating rates.

Japanese yen = 1.00 U.S. dollar



Short-Term Noise vs. Long-Term Equilibrium Path (Exhibit 9.7)

1

The Long-Term Equilibrium Path

Decades of theoretical and empirical studies confirm: exchange rates do adhere to fundamental principles in the long run. There is a fundamental equilibrium path for a currency's value — defined by PPP, BOP fundamentals, and the asset market approach.

2

Short-Term Noise Around the Path

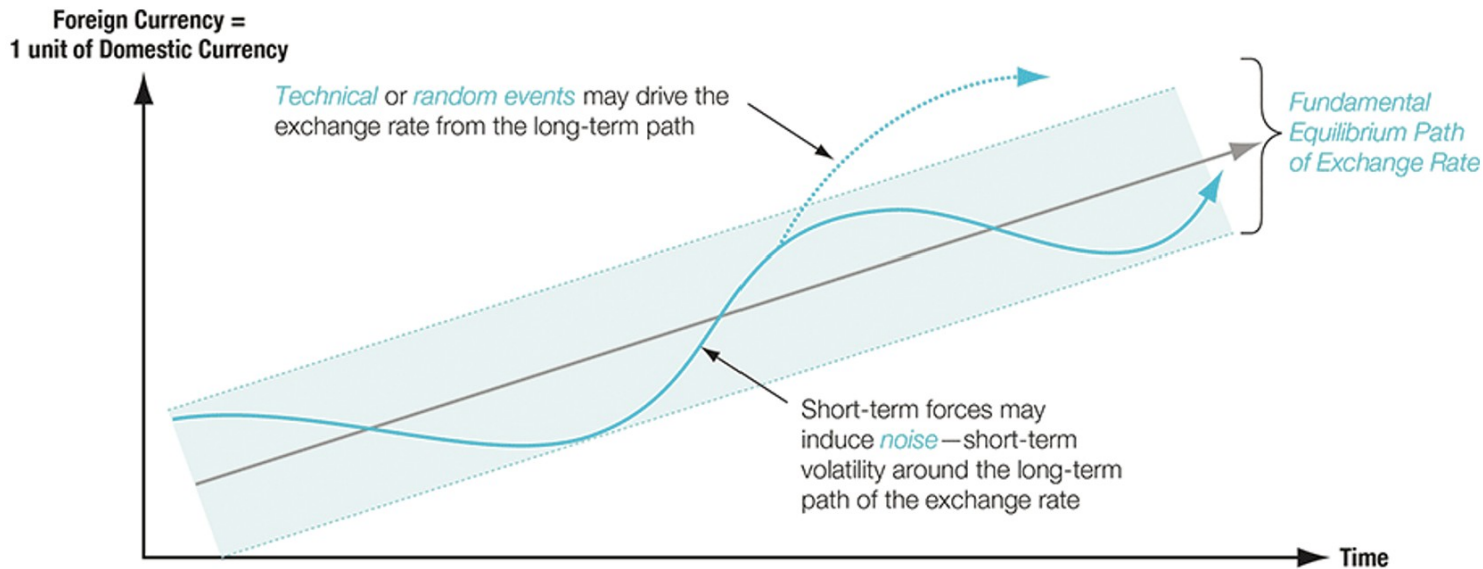
In the short term, random events, institutional frictions, and technical factors may cause currency values to deviate significantly from the long-term path. This is noise. We expect these deviations to occur with some regularity and relative longevity.

Overshooting (Exhibit 9.8):

A currency's value may initially overshoot its new long-run equilibrium after a policy change or shock — moving beyond S_2 before gradually correcting back. Example: after a sudden increase in U.S. money supply, the dollar initially falls below its new long-run equilibrium value S_2 (overshoots to S_1), then slowly appreciates back as short-run technical dominance gives way to long-run fundamental dominance.



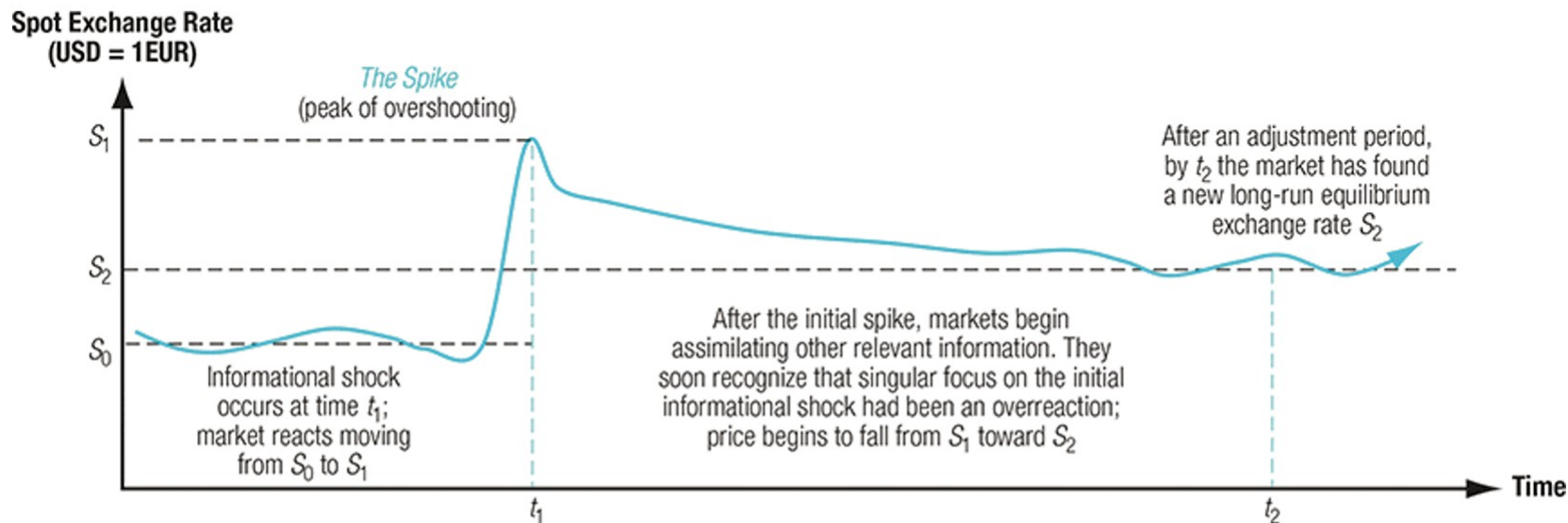
Exhibit 9.7 Short-Term Noise Versus Long-Term Trends



If market participants have *stabilizing expectations*, when forces drive the currency's value below the long-term fundamental equilibrium path, they will buy the currency driving its value back toward the fundamental equilibrium path. If market participants have *destabilizing expectations*, and forces drive the currency's value away from the fundamental path, participants may not move immediately or in significant volume to push the currency's value back toward the fundamental equilibrium path for an extended period of time (or possibly establish a new long-term fundamental path).



Exhibit 9.8 Exchange Rate Dynamics: Overshooting



Which Method for Which Horizon? (Exhibit 9.6)

Period	Regime	Recommended Forecast Methods
SHORT-RUN	Fixed-Rate	1. Assume rate maintained 2. Indicators of stress on fixed rate 3. Capital controls / black market rates 4. Government's capability to maintain peg 5. Changes in official foreign currency reserves
	Floating-Rate	1. Technical methods (trend capture) 2. Forward rates as forecasts: <30 days = random walk; 30-90 days = forward rates 3. 90-360 days: combine trend + fundamental analysis of inflation 4. Government declarations & cooperative agreements on FX goals
LONG-RUN	Fixed-Rate	1. Fundamental analysis 2. BOP management 3. Ability to control domestic inflation 4. Ability to generate hard currency reserves for intervention 5. Ability to run trade surpluses
	Floating-Rate	1. Focus on inflationary fundamentals and PPP 2. Indicators of general economic health (growth and stability) 3. Technical analysis of long-term trends; long-term 'waves' research

Why Governments Intervene: Motivations

Foreign currency intervention = the active management, manipulation, or intervention in the market's valuation of a country's currency. Governments intervene when market exchange rates conflict with their economic or political objectives.



Goal: Fight Inflation

Action →	STRENGTHEN the currency
How →	Buy own currency with FX reserves
Why it works →	Stronger currency → cheaper imports → lower domestic price levels → reduced inflation



Goal: Stimulate Growth (Beggar-thy-neighbor)

Action →	WEAKEN the currency
How →	Sell own currency / print money
Why it works →	Weaker currency → cheaper exports for foreigners → higher export demand → GDP growth

The IMF (Article IV) encourages members to avoid 'currency manipulation' — defined as protracted large-scale intervention in one direction. Many governments quietly ignore this.

Methods of Intervention

1

Direct Intervention

Active buying/selling of the domestic currency against foreign currencies in the open market.

- Buy own currency → strengthen (uses FX reserves — limited by reserve supply)
- Sell own currency → weaken (no physical limit; can theoretically 'print money')
- Single-bank effort may be insufficient — markets may be too large
=>Coordinated Intervention
- Japan intervened 340 days between 1991-2005; avg. effect $\approx 1\text{¥}/\$$

2

Indirect Intervention

Alteration of economic or financial fundamentals that drive capital flows into/out of the currency.

- Raise real interest rates → attract capital → strengthen currency
- Lower real interest rates → capital outflow → weaken currency
- Overly stimulating money supply growth may prove inflationary
- Example: Turkey 2014 raised repo rate from 4.5% to 10% → failed; lira continued to fall

3

Capital Controls

Government restrictions on access to foreign currency or limitations on the ability to exchange domestic for foreign currency.

- Governments may restrict currency access to commercial trade only
- Portfolio and investment transactions often prohibited or limited
- Example: China — daily rate set by government; bureaucratic approval required

Rules of Thumb for Effective Intervention (GFP 9.2)

When a government or central bank wants to influence a currency market, these four guidelines increase the chances of success.



Don't Fight a Strong Trend

Markets moving strongly in one direction are very tough to turn. Intervention during strong market movements will likely result in a very expensive failure. Time interventions carefully — choose moments when trading volumes are light and direction is nearly flat.



Coordinate Timing & Activity

Work together with others and intervene at the right moment — possibly other central banks. Markets react more strongly when they believe multiple institutions support the same goal.



Use Good News

Intervene when positive news supports your message. Investors are more likely to believe a currency will strengthen if good economic news arrives at the same time. E.g. If the government wants a stronger currency, announcing intervention on the same day as strong GDP growth makes the intervention more convincing.



Don't Be Cheap — Overwhelm Them

If you intervene, make it large enough to matter. A small intervention may be ignored by the market and waste money. A large intervention can make traders fear they are on the wrong side of the market and encourage them to reverse their positions.

Disequilibrium: Why Emerging Markets Behave Differently

Large and liquid capital/currency markets follow many of the theories outlined so far in the medium to long term. Smaller and less liquid markets frequently demonstrate behaviors that seemingly contradict theory. The problem lies not in the theory, but in the relevance of the assumptions underlying the theory.



Weak Banking Systems

Banks that are illiquid or insolvent amplify currency shocks. Bank runs accelerate currency collapse. Without functioning banks, the 'plumbing' of business shuts down.



Thin Capital Markets

Small markets can be pushed around by large investors. Large foreign investors can move prices dramatically



Pegged Exchange Rates

Fixed rates invite speculative attack when fundamentals diverge from the peg. The longer the divergence continues, the larger the eventual correction.



Current Account Deficits

Persistent deficits signal currency vulnerability. Countries transitioning from net exporters to net importers become dependent on continued capital inflows.



Corporate Socialism & Governance

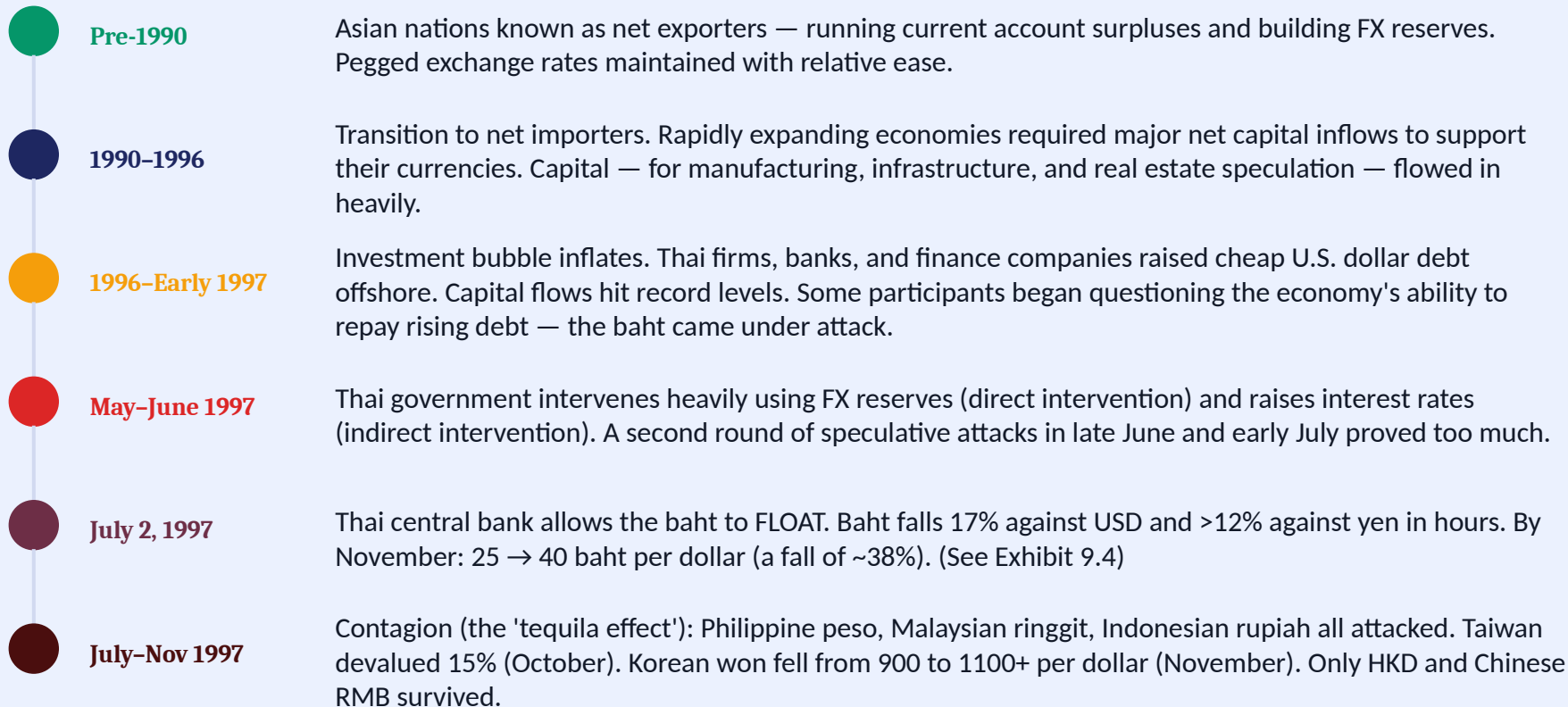
Government implicit guarantees (moral hazard) + cronyism + weak boards erode investor confidence. 'Lifetime employment' and family-controlled conglomerates hide true financial conditions.



Contagion Effects & Speculation

Crisis spreads across borders as investors reprice regional risk. Contagion can cause an 'innocent' country to experience capital flight. Speculation can trigger or worsen any crisis.

The Asian Currency Crisis of 1997: Origins & Trigger



Asian Crisis: Causal Complexities Beyond BOP

The Asian economic crisis — for it was more than just a currency collapse — had many roots besides traditional balance-of-payments difficulties. Three underlying structural similarities across the affected countries:

A Corporate Socialism(Moral Hazard)

For many years, businesses and banks believed that governments would rescue them if they got into trouble. People take greater risks when they believe someone else will bear the cost if things go wrong.

“A company borrows huge amounts of money for risky real estate projects because it believes: "If we fail, the government will save us."

B Cronyism & Poor Corporate Governance

Many firms were controlled either by families or by groups related to the governing party. Business decisions were often influenced by relationships rather than economic merit. Banks sometimes lent money based on connections rather than creditworthiness.

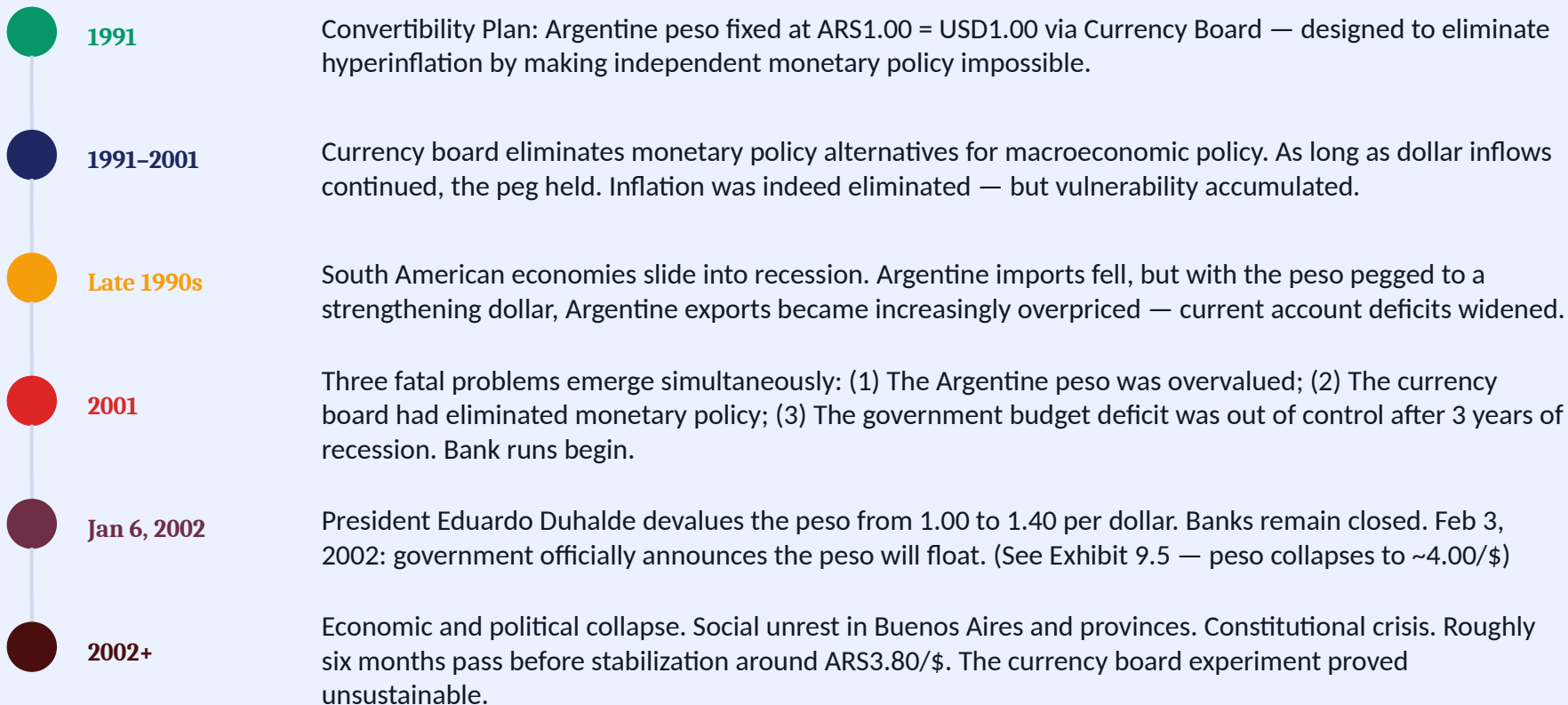
“A bank gives a large loan to a politically connected company even though the company is already losing money. The loan is approved because of personal relationships rather than sound business judgment. When the company fails, the bank suffers losses.

C Banking Sector Collapse

Banks are the "plumbing" of the economy. Just as water cannot flow if pipes break, money cannot flow if banks stop functioning.

“A bank lends money to 100 property developers. If 50 developers go bankrupt, the bank may not recover its money and may also fail.

The Argentine Currency Crisis of 2002: Currency Board & Collapse



Asian (1997) vs. Argentina (2002): Side-by-Side

Dimension	Asian Currency Crisis (1997)	Argentine Crisis (2002)
Exchange Rate Regime	Managed peg (baht.)	Hard peg — Currency Board (ARS1 = USD1)
Primary Trigger	Capital inflows stopped → speculative attack on baht	3 simultaneous problems: overvaluation + no monetary policy + fiscal deficit
Root Structural Causes	Corporate socialism, cronyism, banking fragility	Currency board eliminated monetary flexibility; recession deepened fiscal crisis
Contagion?	Severe — 6+ Asian currencies attacked within months	Mostly contained to Argentina; broader S. American recession
Policy Response	Direct intervention + rate hikes → then floated baht	Devaluation Jan 6, 2002 → full float Feb 3, 2002
Outcome	Region-wide recession; IMF bailouts; structural reforms	Banking system insolvency; constitutional crisis; sovereign default
Lesson	Speculative attacks exploit pegs AND structural weaknesses	Rigid pegs sacrifice monetary policy — works in short term but fragile

Discussion & Review Questions

From Eiteman et al., Chapter 9 End-of-Chapter Questions

Q1

Exchange Rate Determination

What are the three basic theoretical approaches to exchange rate determination? Are they competing or complementary? Why does the textbook combine them?

Q2

PPP Inadequacy

The most widely accepted theory of FX rate determination is PPP, yet it has proven quite poor at forecasting future spot exchange rates. Explain the theoretical and empirical problems with PPP.

Q3

Data and the BOP Approach

Statistics on a country's BOP are used by the business press and business itself, but the academic profession is highly critical of this approach. Why the discrepancy between academic and practitioner views?

Q4

Supply and Demand

Which of the three major theoretical approaches puts the most weight on the supply and demand for currency? What is its primary weakness?

Q5

Asset Market Approach

Explain how the asset market approach can be used to forecast spot exchange rates. How does it differ from the BOP approach to forecasting?

Q6

Intervention Motivation

Why do governments and central banks intervene in the foreign exchange markets? If markets are efficient, why not let them determine the value of a currency?

Discussion & Review Questions

Applied Analysis (continued)

Q7

Direct Intervention Usefulness

When is direct intervention likely to be most successful? When is it likely to be least successful? Apply the 'rules of thumb' from GFP 9.2.

Q9

Asian Crisis 1997

What was the primary disequilibrium at work in Asia in 1997 that likely caused the Asian financial crisis? What structural factors (beyond BOP) made the crisis so severe?

Q11

Forecasting Horizon

You are forecasting USD/VND for: (a) next week, (b) 6 months, (c) 3 years. Use Exhibit 9.6 to select the appropriate method(s) for each horizon and justify your choices.



Capital Controls

Are capital controls really a method of currency market intervention, or more of a denial of activity? How does this fit with the concept of the impossible trinity?

Q10

Currency Crises Comparison

Compare and contrast the currency crises of Russia in 1998 and Argentina in 2002 (Mini-Case: Russian Ruble Roulette). What role did speculation play in each?

Q12

Noise vs. Equilibrium

Using Exhibit 9.7, explain the distinction between short-term noise and the long-term equilibrium path. What are 'stabilizing expectations' vs. 'destabilizing expectations'?

Applying Relative PPP: Step-by-Step

Problem (from textbook example, p. 263):

Current spot rate: ¥90.00 per USD. Japanese prices are expected to increase by 2%, while U.S. prices increase by 1% over the coming period. What does Relative PPP predict for the spot rate next period?

Formula

$$S(t+1) = S(t) \times [(1 + \Delta \text{ in Japanese prices}) / (1 + \Delta \text{ in U.S. prices})]$$

Step 1: Identify inputs

$$S(t) = \text{¥}90.00/\$ \quad | \quad \pi_{\text{Japan}} = 2\% = 0.02 \quad | \quad \pi_{\text{US}} = 1\% = 0.01$$

Step 2: Apply formula

$$S(t+1) = \text{¥}90.00 \times (1.02 / 1.01) = \text{¥}90.00 \times 1.00990 = \text{¥}90.89/\$$$

Answer & Interpretation

PPP predicts the yen will **DEPRECIATE** slightly from ¥90.00 to ¥90.89 per dollar.
The higher-inflation country's currency is expected to weaken.



Key Rule: The higher-inflation country's currency depreciates. Higher inflation = PPP predicts weaker currency over time

Japan's September 2010 Intervention: Analysis

Context: Japan intervened in FX markets for the first time in nearly six years. Japan reportedly bought ~\$20 billion USD in exchange for yen. Finance Ministry officials publicly stated that 82 yen per dollar was 'probably the limit of their tolerance' for yen appreciation.

? What was Japan's goal?

To WEAKEN the yen (slow its appreciation) and protect the competitiveness of Japanese exporters like Toyota and Sony, whose overseas revenues were shrinking in yen terms.

? What type of intervention?

DIRECT intervention: Bank of Japan actively SOLD yen (its own currency) and bought USD in the open market, increasing the supply of yen and reducing upward pressure on the exchange rate.

? Was it sufficient?

Single-bank direct intervention often has limited impact in a \$5+ trillion/day market. History shows (Exhibit 9.3) that Japanese yen selling intervention has rarely achieved a permanent directional change.

? What indirect tool could complement it?

The Bank of Japan could lower real interest rates to reduce the attractiveness of yen-denominated assets, encouraging capital outflows and reducing demand for yen.

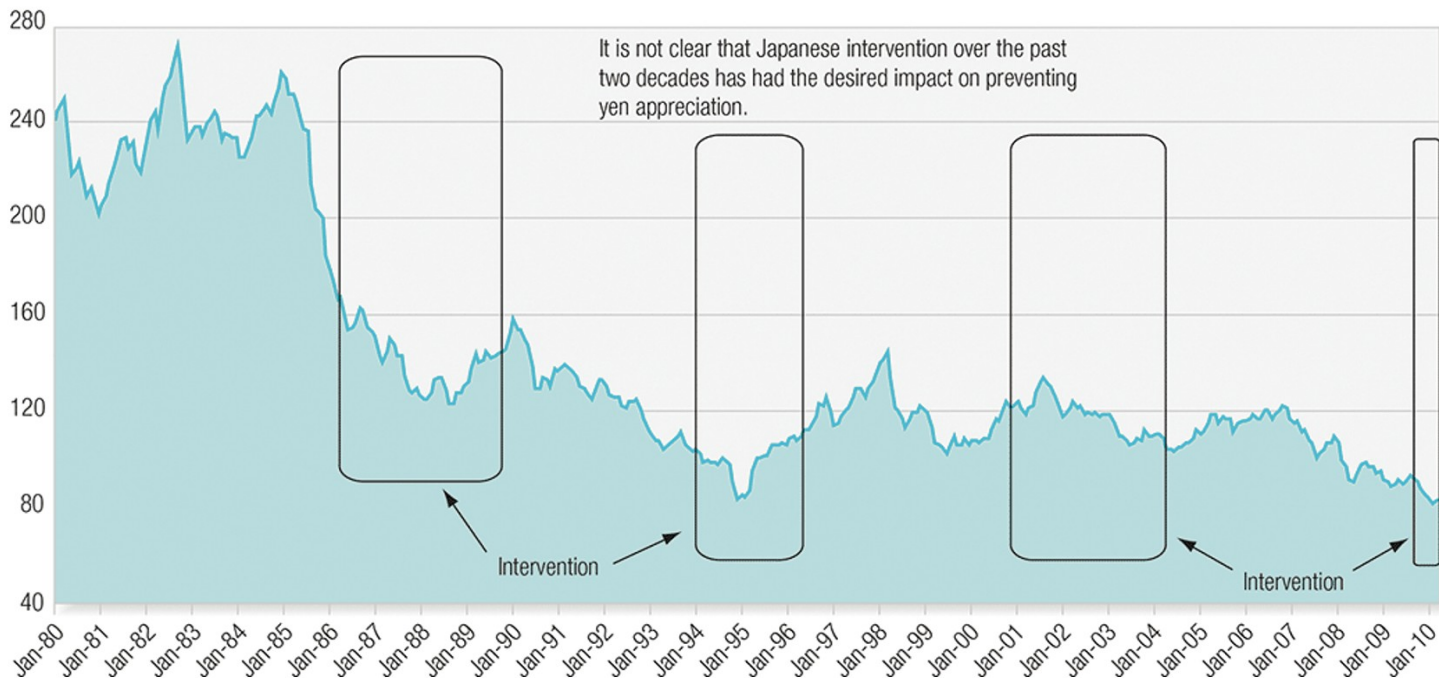
? How does Exhibit 9.3 contextualize this?

The History of Japanese Intervention (Exhibit 9.3) shows intervention episodes since the 1980s — interventions are marked with boxes on the chart. No clear evidence that intervention permanently stopped the long-term yen appreciation trend.



Exhibit 9.3 The History of Japanese Intervention

Japanese yen (JPY) = USD1.00



Chapter 9: Key Summary Points



Intervention Is Always Present

Governments intervene via direct (buy/sell), indirect (interest rates), or capital controls. Even if intervention fails, understanding motivations is critical for exchange rate forecasting. Intervention may have limited effectiveness in large markets.



Emerging Markets Periodically Break Theory

Large, liquid markets follow theory well in the medium-to-long term. Smaller, less liquid markets frequently contradict theory — not because theory is wrong but because its assumptions (banking stability, governance, political safety) are violated.



Short-Term Noise is Real and Persistent

Random events, institutional frictions, and technical factors cause regular deviations from the long-term fundamental equilibrium path (Exhibit 9.7). These deviations can be significant and persist for extended periods.



Currency Crises Have Structural Roots

The Asian (1997) and Argentina (2002) crises illustrate that exchange rate crises are often the trigger, not the full cause. Corporate socialism, governance failure, banking fragility, and contagion turn a currency crisis into an economic catastrophe.

Three Theories. Two Crises. One Framework.

- PPP + BOP + Asset Market = complementary lenses, not competitors
- Central banks intervene via direct action, indirect policy, or capital controls
- Short-term forecasting: technical analysis (noise); Long-term: fundamentals (equilibrium path)
- Emerging market crises are never just about exchange rates — structural factors amplify
- *The herd instinct among forecasters makes sheep look like independent thinkers — Edgar Fiedler*